

RF Admittance level switch

For Liquid Applications Model: RFLS-300L



PRODUCT DESCRIPTION:

The RFLS-300L RF Admittance type level switch is a simple, low cost, rugged point Level Switch designed for Level detection of Liquids, Oils in Tank. This level switch can be used for detecting high, low level or for differential level detection and is used in various level detection applications like Oil, Petrochemicals, slurries, Resins, power utilities industry, food industry, pulp/paper industry, pharmaceuticals industry, water/ treatment plants, foundry industry etc.

The specialty of the probe is that it tends to ignore the build-up on the probe, thus making it suitable for application, which tends to stick on the probe.

The unit has integral electronics along with time delay feature, LED indication, fail safe mode etc.

SALIENT FEATURES:

- Advanced RF Admittance Technology
 - Suitable for Liquid, Oil, Slurry & Chemical Applications
 - Suitable for Conductive & Non-Conductive Liquids
 - Build-up Compensation for reliable level detection
 - No Moving Parts – Long Service Life
 - High / Low Fail-Safe Selection
 - Adjustable Time Delay to prevent false switching
 - LED Level Status Indication
 - Weatherproof / Flameproof Enclosure as per requirement
 - Integral / Remote Electronics available
 - Robust Industrial Construction
 - SS304 / SS316 Probe options available
 - Customised Probe Length & Process Connection
 - Suitable for Top & Side Mounting
 - Easy Installation and Low Maintenance
- Suitable for Sticky & Viscous Liquid Applications



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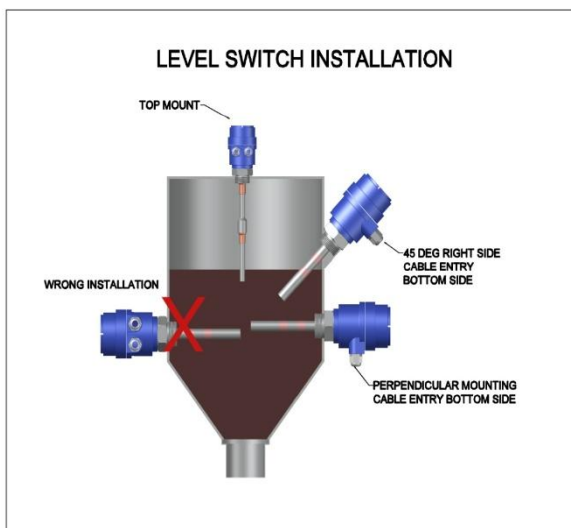
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OPERATING PRINCIPLE:

The RFLS-300L RF Admittance Level Switch operates on the RF Admittance Sensing Principle for efficient, reliable and repeatable point-level detection of liquids. a high-frequency RF signal is applied to the sensing probe. When the probe is surrounded by air, the sensing circuit remains in its normal reference condition. when the liquid reaches the sensing probe, the electrical admittance between the probe and vessel ground changes according to the dielectric properties of the liquid. The electronic circuit continuously detects this change. when the admittance change reaches the preset Sensitivity Threshold, the instrument detects the presence of liquid and operates the Output Relay, providing a level signal for High-Level or Low-Level control. the Active Shield/Guard minimizes the effect of conductive liquid coating or buildup on the sensing probe, reducing false switching and providing stable, efficient and reliable level detection. the RFLS-300L provides Adjustable Sensitivity and Fail-Safe High/Low Selection to suit different liquid and process requirements.

PROBE CONSTRUCTION SUITABLE FOR LIQUID BUILD-UP:



The RFLS-300L a conventional probe may be affected by conductive liquid coating or material build-up, which can cause false switching. The COAT Guard electrode in the RFLS-300L uses special Build-Up Sensing Technology to minimize the effect of coating on the sensing probe.

The COAT Guard electrode is electrically insulated from the active sensing probe and is driven with the same RF frequency and polarity as the active electrode. When conductive liquid coating or build-up forms on the probe, the COAT Guard circuit effectively saturates the build-up region, preventing unwanted RF current flow from the active probe to the vessel wall.

When the liquid level reaches the sensing probe, the change in RF Admittance is detected above the saturated build-up region, allowing the electronic circuit to operate the output reliably. This construction significantly reduces false switching due to conductive liquid coating or build-up and provides stable, reliable and repeatable level detection.

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COMPONENTS OF COMPACT LEVEL SWITCH ARE EXPLAINED IN FIGURE

ELECTRICAL CONNECTION
UNIVERSAL /230 VAC/110VAC/24 VDC

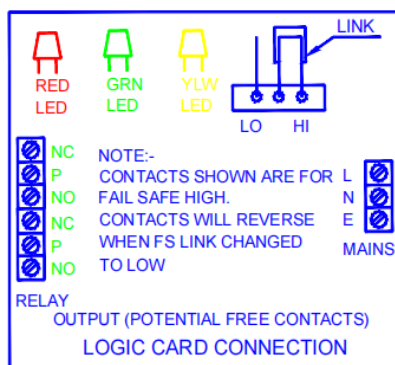
Integral Type



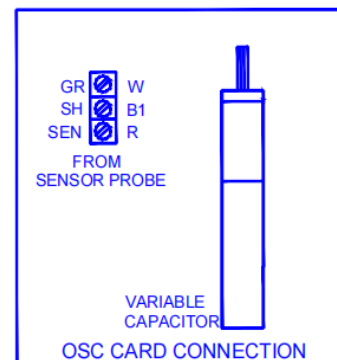
Remote Type



Controller /RF Cable/ Remote Probe



Controller Card



Oscillator Card

DATA SHEET

RFLS-300 – RF Admittance Level Switch
Selection / Ordering Information

Selection	Code	Description
Model No.	RFLS-300	RF Admittance Level Switch
Application	L	Liquid
Mounting Type	IN/RE	Integral / Remote
Probe Type	RD / RP	Rigid Rod / Rope Flexible
Insulation	FP/PT/ CE	Full PTFE / Part PTFE / Ceramic
Probe Diameter	D1 / D2	Standard 16mm / Heavy Duty 30mm
Probe Material	04 /4L/ 16/ 6L/MS/ SM	SS304/SS304L / SS316/ SS316L /Mild steel/ Special material
Power Supply	AC/DC/A1/UN	230VAC / 24VDC / 110VAC / Universal
Output	DP/SP/PN	DPDT Relay /SPDT Relay / PNP Relay
Switching Delay	ND / DA	No Delay / Adjustable
Indication	L / D	LED / Digital Display
Enclosure	WP / FL	Weatherproof / Flameproof
Connection	T	Threaded
	F	Flanged
Mounting	B1 / B2 / B3/B4	1" / 1½" / 2" BSP/ Special Mounting
	N1 / N2 / N3/N4	1" / 1½" / 2" NPT/ Special Mounting
Flange Rating	A1 / A3 / A6	ANSI 150# / ANSI 300# /ANSI 600#
Flange Facing	RF / FF	Raised Face / Flat Face

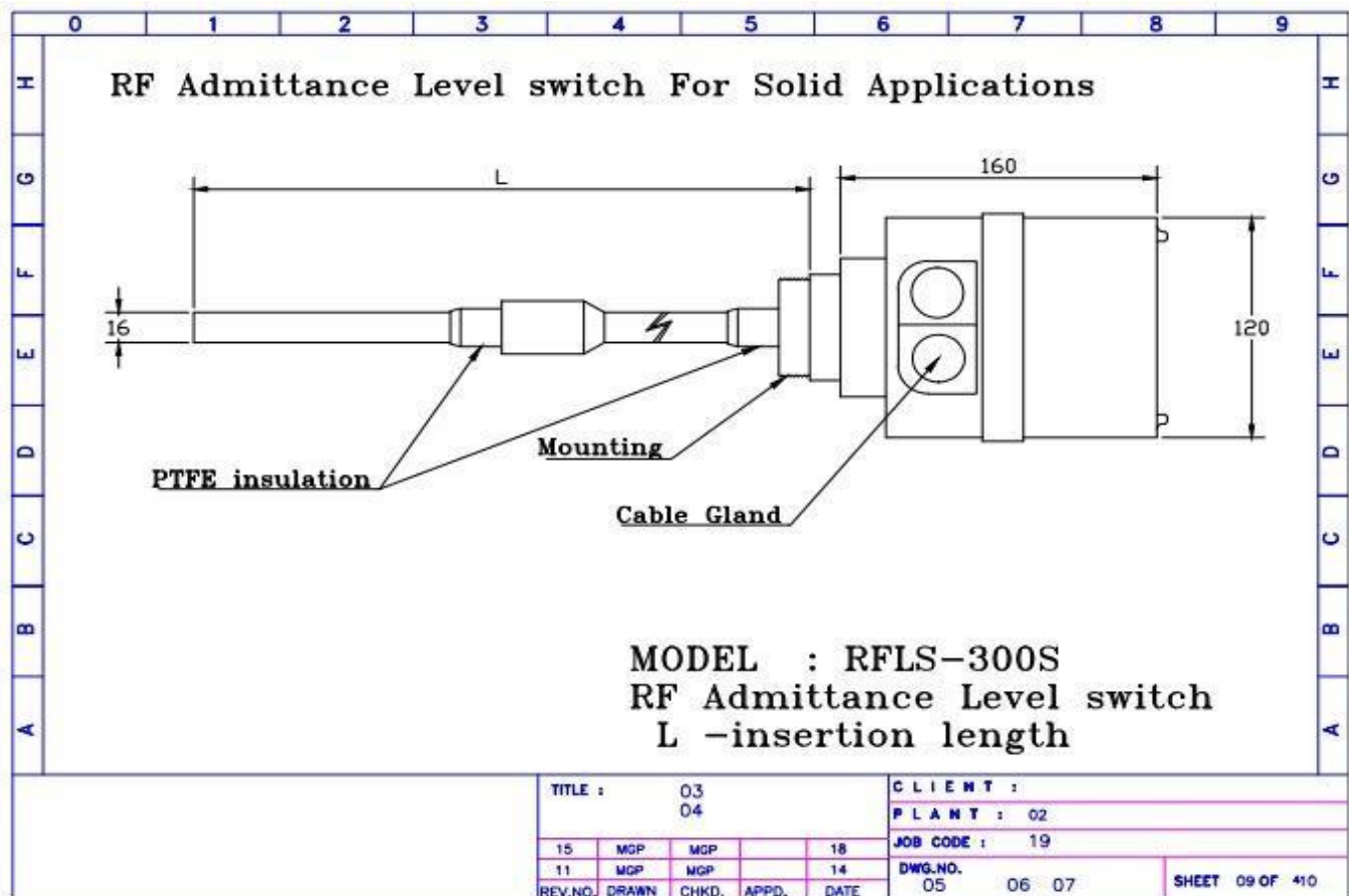


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Selection	Code	Description
MOC-Process Connection	S4 / S6/ S16/MS/SL	SS304 / SS316 /SS316L /Mild steel / Special material
Flange Size	01 / 15 / 02 / 03 / 04/PS	1" / 1½" / 2" / 3" / 4"/ Special Process
Pressure	P1 /P2/P3/P4	Standard 0 to 2KG/ 2 To 10kg /10kg to 40kg / Customized
Temperature	T1/ T2/TS	Max -20–150°C / 200°C/ Customer Specified Special Designed
High Temperature stand off	H8/H1/H2/H3/HS	80MM/150 MM /200MM /300 MM /Special standoff Extension
Remote Cable	05 / 10 / 15 / 20 / 25 / 30/SR	5 / 10 / 15 / 20 / 25 / 30 m/ Special Customer Requirement
Cable Gland	G1/ G2 / G3/ G4	PG13.5 / ½"NPT / ¾" ET/ PG16
Ground / Reference	GS / GX	Standard / Special
Material Certificate	M0 / MC	Not Required / Material test certificate
Test Certificate	TC / CC	Test Certificate / Calibration Certificate
Documentation	DS / GA / IR	Datasheet / GA / Inspection Report
Tagging	TAG	Customer Tag No.
Probe Length	300	mm

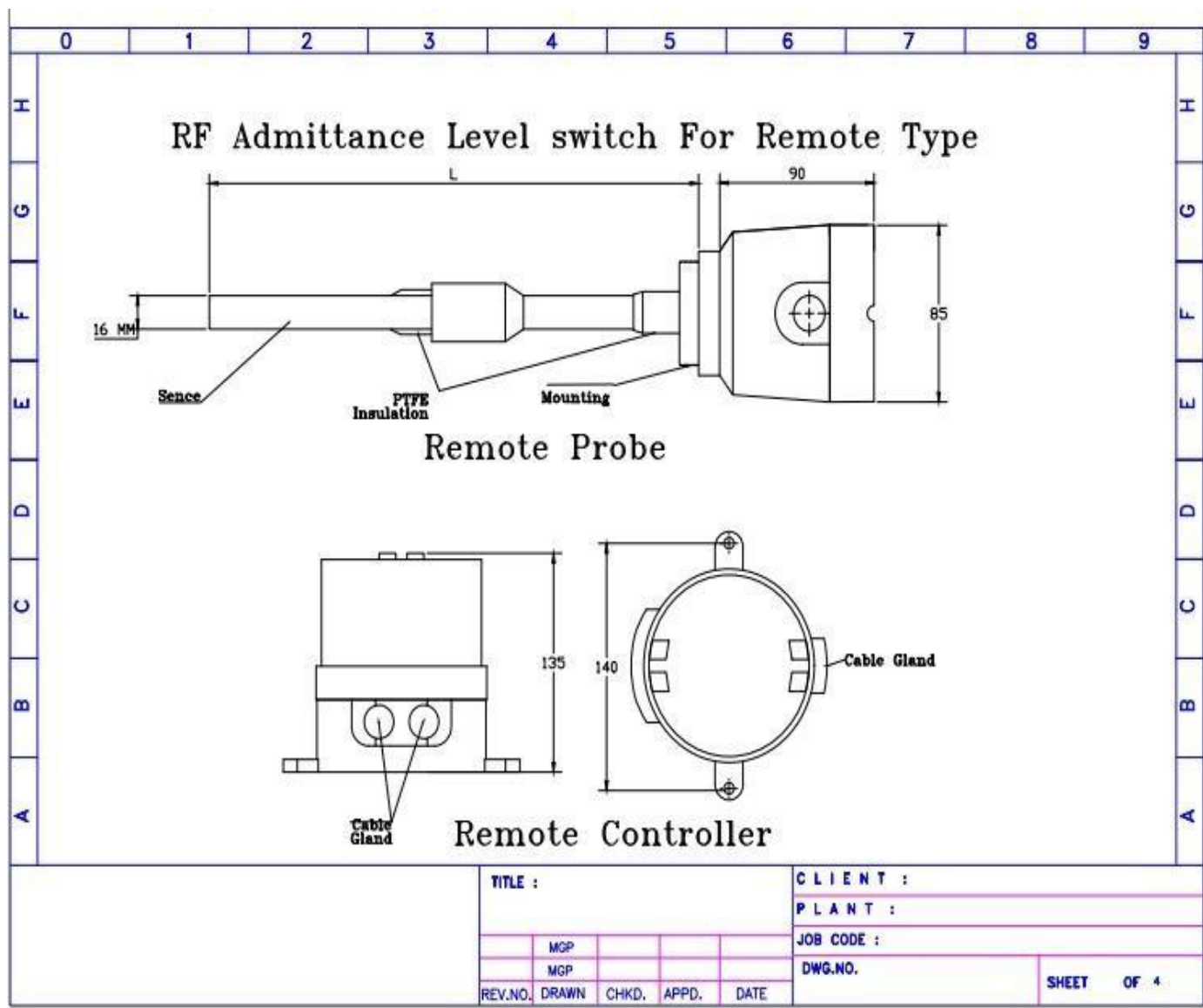
Mechanical Drawing: Integral Type:



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Mechanical Drawing: Remote Type:



Manufactured By:

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